

ZHENHUA XU

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Position

Tsinghua University

Postdoc (Shuimu Scholar) in School of Vehicle and Mobility, work with [Prof. Jianqiang Wang](#)

Sep. 2024 – Current

Beijing, China

The University of Hong Kong

Postdoc, work with [Prof. Hengshuang Zhao](#)

Apr. 2023 – Jul. 2024

HKSAR, China

Education

Hong Kong University of Science and Technology

Ph.D. of CS. **HKPFS** awardee. Supervised by [Prof. Ming Liu](#) and [Prof. Huamin Qu](#)

Sep. 2018 – Mar. 2023

HKSAR, China

University of California, Los Angeles

CSST (Cross-disciplinary Scholars in Science and Technology) program (GPA 4.0/4.0)

Jul. 2017 – Sep. 2017

LA, USA

Harbin Institute of Technology

Bachelor in Electronics and Information Engineering (Score 91.19/100, **ranking 1/88**)

Sep. 2014 – Jun. 2018

Harbin, China

Zhengzhou Foreign Language School

Sep. 2011 – Jun. 2014

Zhengzhou, China

Research Keywords

Autonomous Driving, Large Language Model, High-definition Map, Robotics, Computer Vision

Research Experience

Large Language Model for Autonomous Driving | *Robotics, Computer vision*

2023-Current

- Autonomous vehicles promise reduced traffic, fewer accidents, and increased sustainability. However, a key challenge is achieving explainable autonomous driving, where AI systems transparently justify decisions—essential for commercialization and addressing safety, ethical, and legal concerns.
- The emergence of large language models (LLMs, e.g., ChatGPT, GPT4) enhances explainable autonomous driving by translating technical data into everyday language, providing context-aware AI explanations, and augmenting training. They also address ethical and legal concerns, fostering trust and seamless human-AI interactions.
- We propose leveraging large language models (LLMs) to demystify autonomous driving. LLMs are trained to control vehicles like humans and offer conceptual explanations for each specific action, enhancing transparency and understanding.
- Several research academic research projects are ongoing.

Vector Map Creation | *Robotics, Computer vision*

2019-2024

- Vector maps, including standard and high-definition maps, are crucial for autonomous vehicles as they supply navigation and planning algorithms with vital information about static line-shaped objects like road boundaries and networks. However, manual creation is inefficient and labor-intensive. Thus, there is a need for efficient and effective methods to automatically generate vector maps of target objects.
- Our team proposed to handle this problem from multiple aspects, such as offline global map construction by imitation learning, online map detection by transformers, and graph manipulation. All projects achieved promising results.
- Multiple papers including 8 first-author papers are accepted by top journals and conferences in the robotics communities (e.g., RA-L, ICRA, IROS, ECCV, TGRS).

Preprints(*Equal contribution, †Corresponding author)

- [1] Z. Wang, Y. Jiang, R. Zhou, B. Zhang, F. Zhang[†], **Z. Xu[†]**, Y. Zhang, J. Wang. "DriveCode: Domain Specific Numerical Encoding for LLM-Based Autonomous Driving." arXiv preprint arXiv:2603.00919, 2026.
- [2] Y. Wang, H. Huang, **Z. Xu**, K. Sun, B. Guo, J. Zhao. "Learning from Risk: LLM-Guided Generation of Safety-Critical Scenarios with Prior Knowledge." arXiv preprint arXiv:2511.20726, 2025.
- [3] Z. Li, J. Yang, Y. Zhao, L. Ren, X. Wu, **Z. Xu**, H. Zhao. "Sim-to-Real Dynamic Object Manipulation on Conveyor Systems via Optimization Path Shaping." Under review, 2025.

Full List of Publications(*Equal contribution, [†]Corresponding author)

- [1] Z. Ke, Z. Liu, Y. Wang[†], J. Li, J. Jiang, Y. Jiang, **Z. Xu**[†], J. Wang. "GRADE: Guiding Realistic Autonomous Driving with Adaptive Trajectory Evolution." in The Conference on Computer Vision and Pattern Recognition Fundings (CVPR Fundings), 2026.
- [2] **Z. Xu**, Y. Bai, Y. Zhang, Z. Li, F. Xia, K.K. Wong, J. Wang and H. Zhao. "DriveGPT4-V2: Harnessing Large Language Model Capabilities for Enhanced Closed-Loop Autonomous Driving." in The Conference on Computer Vision and Pattern Recognition (CVPR), 2025. (**Highlight Paper**)
- [3] B. Zhang, H. Huang, C. Liu, Y. Zhang, **Z. Xu**[†]. "InVDriver: Intra-Instance Aware Vectorized Query-Based Autonomous Driving Transformer," Journal of Intelligent and Connected Vehicles (JICV), 2025.
- [4] Z. Li, X. Xu, **Z. Xu**, S. Lim, H. Zhao. "LARM: Large Auto-Regressive Model for Long-Horizon Embodied Intelligence.", International Conference on Machine Learning (ICML), 2025.  [Web Page](#)
- [5] Z. Li, L. Ren, J. Yang, Y. Zhao, X. Wu, **Z. Xu**, X. Bai, H. Zhao. "VIRT: Vision Instructed Transformer for Robotic Manipulation.", International Conference on Machine Learning (ICML), 2025.
- [6] **Z. Xu**, Y. Zhang, E. Xie, Z. Zhao, Y. Guo, K.K. Wong, Z. Li and H. Zhao. "DriveGPT4: Interpretable End-to-end Autonomous Driving via Large Language Model." in IEEE Robotics and Automation Letters (RAL), 2024.  [Web Page](#) (**ESI Highly Cited Paper**)
- [7] **Z. Xu**, K.K. Wong and H. Zhao. "InsMapper: Exploring Inner-instance Information for Vectorized HD Mapping." in The European Conference on Computer Vision (ECCV), 2024.  [Web Page](#)
- [8] H. Liu, L. Zheng, X. Yan, B. Xue, **Z. Xu**, and M. Liu, "PGO-IPM: Enhance IPM Accuracy with Pose-guided Optimization for Low-cost High-definition Angular Marking Map Generation," in IEEE Intelligent Vehicles Symposium (IV), 2024.
- [9] **Z. Xu**, Y. Liu, Y. Sun, L. Wang, and M. Liu, "RNGDet++: Road Network Graph Detection by Transformer with Instance Segmentation and Multi-scale Features Enhancement," in IEEE Robotics and Automation Letters (RAL), 2023.  [Web Page](#)
- [10] **Z. Xu**, Y. Liu, Y. Sun, L. Wang, and M. Liu, "CenterLineDet: CenterLine Graph Detection for Road Lanes with Vehicle-mounted Sensors by Transformer for HD Map Generation," in 2023 IEEE/RSJ International Conference on Robotics and Automation (ICRA), 2023.  [Web Page](#)
- [11] Y. Liu, **Z. Xu**, H. Huang, L. Wang, and M. Liu, "FSNet: Redesign Self-Supervised MonoDepth for Full-Scale Depth Prediction for Autonomous Driving," IEEE Transactions on Automation Science and Engineering (TASE), 2023.
- [12] **Z. Xu**, Y. Liu, L. Gan, Y. Sun, L. Wang, and M. Liu, "RNGDet: Road Network Graph Detection by Transformer," in IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2022.  [Web Page](#)
- [13] **Z. Xu**, Y. Liu, L. Gan, X. Hu, Y. Sun, L. Wang, and M. Liu, "csBoundary: City-scale Road-boundary Detection in Aerial Images for High-definition Maps," in IEEE Robotics and Automation Letters (RAL), 2022.  [Web Page](#)
- [14] Y. Liu, **Z. Xu**, and M. Liu, "Star-Convolution for Image-Based 3D Object Detection," in 2022 IEEE/RSJ International Conference on Robotics and Automation (ICRA), 2022.
- [15] **Z. Xu**, Y. Sun, L. Wang, and M. Liu, "CP-loss: Connectivity-preserving Loss for Road Curb Detection in Autonomous Driving with Aerial Images," in 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021.  [Web Page](#)
- [16] **Z. Xu**, Y. Sun, and M. Liu, "Topo-Boundary: A Benchmark Dataset on Topological Road-Boundary Detection Using Aerial Images for Autonomous Driving," in IEEE Robotics and Automation Letters (RAL), 2021.  [Web Page](#)
- [17] **Z. Xu**, Y. Sun, and M. Liu, "iCurb: Imitation Learning-Based Detection of Road Curbs Using Aerial Images for Autonomous Driving," in IEEE Robotics and Automation Letters (RAL), 2021.  [Web Page](#)

- [18] T. Liu*, Q. Liao*, L. Gan, F. Ma, J. Cheng, X. Xie, Z. Wang, Y. Chen, Y. Zhu, S. Zhang, Z. Chen, Y. Liu, M. Xie, Y. Yu, Z. Guo, G. Li, P. Yuan, D. Han, Y. Chen, H. Ye, J. Jiao, P. Yun, **Z. Xu**, H. Wang, H. Huang, S. Wang, P. Cai, Y. Sun, Y. Liu, L. Wang, and M. Liu, "The Role of the Hercules Autonomous Vehicle During the COVID-19 Pandemic: An Autonomous Logistic Vehicle for Contactless Goods Transportation," in IEEE Robotics and Automation Magazine (RAM), 2021.
- [19] Q. Wang, **Z. Xu**, Z. Chen, Y. Wang, S. Liu and H. Qu, "Visual Analysis of Discrimination in Machine Learning," in IEEE Transactions on Visualization and Computer Graphics (TVCG), 2021.
- [20] Y. Zhang, S. Yang, H. Li, **Z. Xu**. "Shadow tracking of moving target based on CNN for video SAR system." in IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium. IEEE, 2018.
- [21] **Z. Xu**, Y. Zhang, H. Li, H. Mu, Y. Zhuang. "A new shadow tracking method to locate the moving target in SAR imagery based on KCF." in International Conference in Communications, Signal Processing, and Systems (CSPS). Springer, Singapore, 2017.

Awards and Honors

- **2024-2026** Shuimu Tsinghua Scholar
- **2018-2022** HKPF (Hong Kong PhD Fellowship)
- **2018** Outstanding Graduate of Harbin Institute of Technology
- **2018** Guanghua Scholarship
- **2017** UCLA CSST (Cross-disciplinary Scholars in Science and Technology)
- **2017** National Scholarship
- **2016** Meritorious Winner in MCM/ICM
- **2014-2018** Renmin Scholarship
- **2014-2018** University Merit Student
- **2013** Provincial 1st prize in National High School Mathematics League (NO.49)

Academic services

- **Organizing committee:**
 - 2024 ICML Workshop, Multi-modal Foundation Model meets Embodied AI (MFM-EAI)
- **Reviewer:**
 - IEEE Robotics and Automation Letters (RA-L)
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
 - IEEE/RSJ International Conference on Robotics and Automation (ICRA)
 - Conference on Computer Vision and Pattern Recognition (CVPR)
 - Conference on Neural Information Processing Systems (NIPS)
 - AAAI Conference on Artificial Intelligence (AAAI)
 - IEEE Transactions on Geoscience and Remote Sensing (TGRS)
 - International Journal of Computer Vision (IJCV)
 - The British Machine Vision Conference (BMVC)
 - Autonomous Vehicle Vision (AAV)
- **Teaching assistant:**
 - COMP3711 (Design and Analysis of Algorithms)
 - COMP3311 (Database Management Systems)